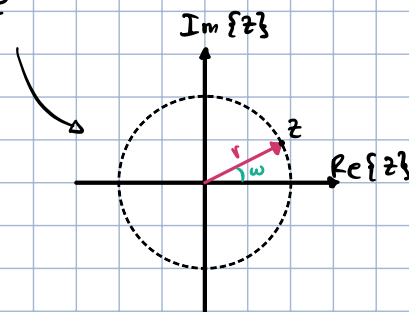


THE Z TRANSFORM IS A GENERALIZATION OF THE DFT, IT'S THE DISCRETE COUNTERPART OF THE LAPLACE TRANSFORM.

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n}, \quad z = r e^{j\omega}$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} r^{-n}$$



THE ADDITION OF \$r^{-n}\$ MEANS THAT, IN ADDITION TO SINES AND COSINES, WE ARE NOW ALSO EVALUATING THE CORRELATION WITH EXPONENTIALS (\$\rightarrow\$ CONVERGENCE PROBLEMS)

IF \$r=1\$ IT IS THE DFT \$\rightarrow\$ Z TRANSFORM ON UNIT CIRCLE

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = \sum_{n=-\infty}^{+\infty} x[n] r^{-n} e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n} = X(e^{j\omega})$$

EX

$$x[n] = \delta[n] \rightarrow X(z) = \sum_{n=-\infty}^{+\infty} \delta[n] z^{-n} = 1$$

$$x[n] = \delta[n-n_0] \rightarrow X(z) = \sum_{n=-\infty}^{+\infty} \delta[n-n_0] z^{-n} = z^{-n_0}$$

$$h[n] = \begin{cases} 1 & 0 \leq n \leq 4 \\ 0 & \text{ELSE} \end{cases}$$

$$\rightarrow H(z) = \sum_{n=-\infty}^{+\infty} h[n] z^{-n} = \sum_{n=0}^4 z^{-n} = z^0 + z^{-1} + z^{-2} + z^{-3} + z^{-4} = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

ALTERNATIVE WAY \$\rightarrow\$

$$H(z) = \sum_{n=-\infty}^{+\infty} (\mu[n] - \mu[n-5]) z^{-n} = \sum_{n=0}^4 (\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] + \delta[n-4]) z^{-n}$$

$$= z^0 + z^{-1} + z^{-2} + z^{-3} + z^{-4} = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4}$$

$$x[n] = \mu[n] \rightarrow X(z) = \sum_{n=-\infty}^{+\infty} \mu[n] z^{-n} = \sum_{n=0}^{\infty} z^{-n} = \frac{z^0 - z^{-\infty}}{1 - z^{-1}} = \frac{1}{1 - z^{-1}} = \frac{z}{z-1}, \quad |z^{-1}| < 1$$

REGION OF CONVERGENCE

THERE IS A REGION OF CONVERGENCE ASSOCIATED TO THE TRANSFORM. WE NEED TO VERIFY CONVERGENCE IN 2 DIMENSIONS.

$$|X(z)| = \left| \sum_{n=-\infty}^{+\infty} x[n] z^{-n} \right| \leq \sum_{n=-\infty}^{+\infty} |x[n]| |z^{-n}| = \sum_{n=-\infty}^{+\infty} |x[n]| r^{-n} < \infty$$

SO FOR THE Z TRANSFORM TO BE PROPERLY DEFINED WE NEED BOTH AN EXPRESSION AND A ROC.

ROCS ARE ALWAYS RINGS CENTERED IN THE ORIGIN

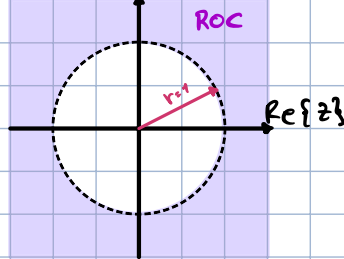
EX

$$h[n] = \mu[n] \rightarrow H(z) = \sum_{n=-\infty}^{+\infty} \mu[n] z^{-n} = \sum_{n=0}^{\infty} z^{-n} = \frac{(z^{-1})^0 - (z^{-1})^{\infty}}{1 - z^{-1}} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \Rightarrow r > 1$$

CONVERGES IF \$|z| > 1\$

OUTSIDE OF UNIT CIRCLE

\$\text{Im}\{z\}\$

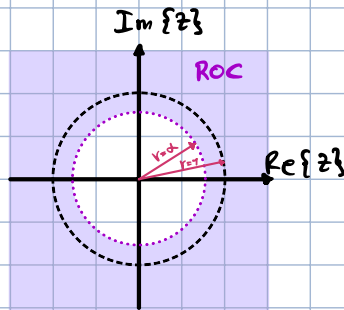


EX

$$h[n] = a^n u[n]$$

$$H(z) = \sum_{n=-\infty}^{+\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (a z^{-1})^n = \left(\frac{a}{z}\right)^0 - \left(\frac{a}{z}\right)^{\infty} = \frac{1}{1 - a z^{-1}}, \quad \left|\frac{a}{z}\right| < 1 \Rightarrow |z| > |a|$$

DTFT ALSO CONVERGES IF $|a| < 1$, SINCE $|z|$ WOULD INCLUDE THE UNIT CIRCLE $r=1$



EX

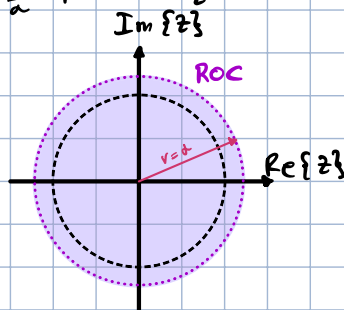
$$h[n] = -a^n u[-n-1]$$

ADD AND REMOVE ZERO TO USE GEOMETRIC FORM

$$H(z) = \sum_{n=-\infty}^{+\infty} -a^n u[-n-1] z^{-n} = \sum_{n=-\infty}^{-1} -a^n z^{-n} = - \sum_{m=1}^{\infty} a^{-m} z^m = - \sum_{m=2}^{\infty} a^{-m} z^m + \sum_{m=0}^0 a^{-m} z^m = - \sum_{m=2}^{\infty} a^{-m} z^m + 1$$

$$= - \frac{\left(\frac{z}{a}\right)^2 - \left(\frac{z}{a}\right)^{\infty}}{1 - \frac{z}{a}} + 1 = - \frac{1}{1 - \frac{z}{a}} + 1, \quad \left|\frac{z}{a}\right| < 1$$

$$\frac{-1 + 1 - \frac{z}{a}}{1 - \frac{z}{a}} = - \frac{\frac{z}{a}}{1 - \frac{z}{a}} = \frac{\frac{z}{a}}{\frac{z}{a} - 1} = \frac{1}{1 - \frac{a}{z}}, \quad \left|\frac{a}{z}\right| > 1 \Leftrightarrow |z| < |a|$$



DTFT CONVERGES IF $a \geq 1$

$a^n u[n]$ AND $-a^n u[-n-1]$ HAVE THE SAME TRANSFORM WITH DIFFERENT ROCs

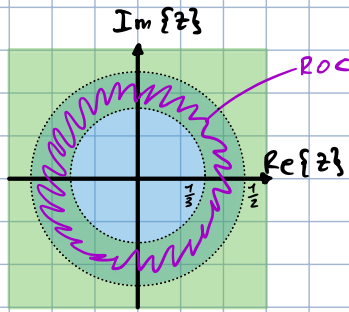
EX

$$x[n] = \left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1]$$

$$X(z) = \sum_{n=-\infty}^{+\infty} \left[\left(-\frac{1}{3}\right)^n u[n] - \left(\frac{1}{2}\right)^n u[-n-1] \right] z^{-n} = \sum_{n=0}^{\infty} \left(-\frac{1}{3}\right)^n z^{-n} - \sum_{m=-\infty}^{-1} \left(\frac{1}{2}\right)^m z^{-m} = \sum_{n=0}^{\infty} \left(-\frac{1}{3z}\right)^n - \sum_{m=1}^{\infty} (2z)^m =$$

$$= \sum_{n=0}^{\infty} \left(-\frac{1}{3z}\right)^n - \sum_{m=0}^{\infty} (2z)^m + \sum_{m=0}^0 (2z)^m = \sum_{n=0}^{\infty} \left(-\frac{1}{3z}\right)^n - \sum_{m=0}^{\infty} (2z)^m + 1 = \frac{1}{1 + \frac{1}{3z}} - \frac{1}{1 - 2z} + 1, \quad |z| > \frac{1}{3} \cap |z| < \frac{1}{2}$$

$$= \frac{1}{1 + \frac{1}{3z}} + \frac{-1 + 1 - 2z}{1 - 2z} = \frac{z}{z + \frac{1}{3}} + \frac{z}{z - \frac{1}{2}}, \quad \frac{1}{3} < |z| < \frac{1}{2}$$



CAUSALITY

THE ROC DESCRIBES ALSO THE CAUSALITY OF A SEQUENCE.

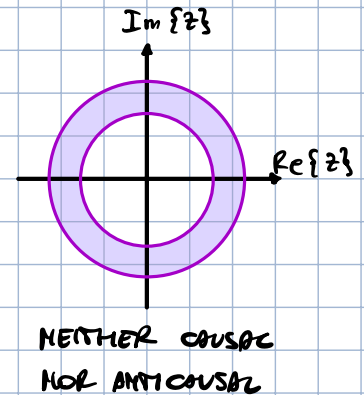
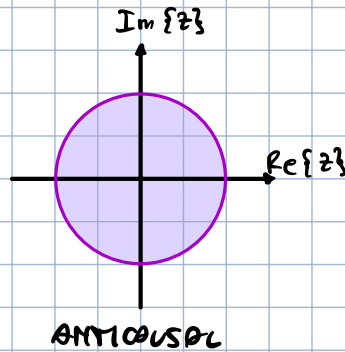
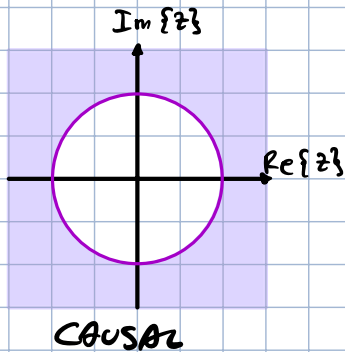
INFINITE DURATION SIGNALS

- $\neq \emptyset$ ONLY FOR $n \geq 0 \rightarrow$ RIGHT SIDED
CAUSAL SEQUENCES MOVE ROCs OUTSIDE OF A RING
- $\neq \emptyset$ ONLY FOR $n \leq 0 \rightarrow$ LEFT SIDED
ANTI-CAUSAL " " " " INSIDE OF A RING
- TWO-SIDED, NON CAUSAL
NEITHER CAUSAL NOR ANTI-CAUSAL " " " AS A RING

$$u[n] \xrightarrow{z} \frac{z}{z-1}, \quad |z| > 1$$

$$u[-n-1] \xrightarrow{z} \frac{z}{z-1}, \quad |z| < 1$$

$$u[n-1] + u[-n-5]$$



THIS IS TRUE ONLY FOR INFINITE DURATION SIGNALS.

FINITE DURATION SIGNALS

IF THE SIGNALS ARE FINITE, THE ONLY CRITICAL POINTS MAY BE $z = \emptyset$ AND $z = \infty$, BECAUSE SINCE THE SIGNAL HAS A FINITE NUMBER OF SAMPLES, IT CAN'T DIVERGE

- CAUSAL: ONLY PROBLEM IN $z = \emptyset$
- ANTICAUSAL: ONLY PROBLEM IN $z = \infty$
- NON CAUSAL: PROBLEM IN BOTH $z = \emptyset$ AND $z = \infty$

PROPERTIES

LINEARITY

$$\left. \begin{aligned} x_1[n] &\longleftrightarrow X_1(z) \\ x_2[n] &\longleftrightarrow X_2(z) \end{aligned} \right\} x_1[n] + x_2[n] \longleftrightarrow X_1(z) + X_2(z)$$

EX

$$x[n] = \left(\frac{1}{2}\right)^n + 2 \mu[n]$$

WE KNOW THAT

$$\mu[n] \longleftrightarrow \frac{z}{z-1}, \quad |z| > 1$$

$$\alpha^n \mu[n] \longleftrightarrow \frac{z}{z-\alpha}, \quad |z| > \alpha$$

THEREFORE THANKS TO LINEARITY

$$x[n] = \left(\frac{1}{2}\right)^n \mu[n] + 2 \mu[n] \longleftrightarrow X(z) = \frac{z}{z-1/2} + 2 \frac{z}{z-1}, \quad |z| > \frac{1}{2} \cap |z| > 1 \Rightarrow |z| > 1$$

WHICH IS QUITE HANDY: IF WE CAN DECOMPOSE A SEQUENCE IN SUB-SEQUENCES FOR WHICH THE TRANSFORM IS KNOWN, WE CAN USE LINEARITY AND AVOID THE CALCULATIONS

TIME SHIFTING

$$x[n] \longleftrightarrow X(z) \longrightarrow x[n-n_0] \longleftrightarrow X(z) z^{-n_0}$$

EX

$$x[n] = \alpha^{n-1} \mu[n-1] \longleftrightarrow \underbrace{\frac{z}{z-\alpha}}_{\mathcal{Z}\{ \alpha^n \mu[n] \}} z^{-1} = \frac{1}{z-\alpha}, \quad |z| > |\alpha|$$

EX

$$x[n] = \delta[n] \longleftrightarrow X(z) = 1$$

$$x[n-n_0] = \delta[n-n_0] \longleftrightarrow X(z) = z^{-n_0} \begin{cases} z \neq 0, & n_0 > 0 \\ |z| \neq \infty, & n_0 < 0 \end{cases}$$

EXPONENTIAL MULTIPLICATION

$$x[n] \longleftrightarrow X(z) \longrightarrow z_0^n x[n] \longleftrightarrow X\left(\frac{z}{z_0}\right)$$

! THE ROC CHANGES!

$$y[n] = z_0^n x[n], \quad |z| \geq r_x \longleftrightarrow Y(z) = \sum_{n=-\infty}^{+\infty} z_0^n x[n] z^{-n} = \sum_{n=-\infty}^{+\infty} x[n] \left(\frac{z}{z_0}\right)^{-n} = X\left(\frac{z}{z_0}\right), \quad |z| \geq |z_0| r_x$$

COMPRESSES/STRETCHES THE ROC

IF A FUNCTION DOES NOT ALLOW TO COMPUTE $\mathcal{Z}$, COMPRESS/EXPAND ROC AND THEN WITH CHANGE OF VARIABLES GO BACK TO FUNCTION OF INTEREST

EX

$$x[n] = r^n \cos(\omega_0 n) \mu[n] = r^n \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2} \mu[n] = \frac{1}{2} (r e^{j\omega_0})^n \mu[n] + \frac{1}{2} (r e^{-j\omega_0})^n \mu[n]$$

$$X(z) = \sum_{n=0}^{+\infty} \left\{ \frac{1}{2} (r e^{j\omega_0})^n + \frac{1}{2} (r e^{-j\omega_0})^n \right\} z^{-n} = \frac{1}{2} \left[\sum_{n=0}^{+\infty} \left(\frac{r e^{j\omega_0}}{z} \right)^n + \sum_{n=0}^{+\infty} \left(\frac{r e^{-j\omega_0}}{z} \right)^n \right] =$$

$$\left| \frac{r e^{j\omega_0}}{z} \right| < 1 \rightarrow \left| \frac{r}{z} \right| < 1, |z| > |r|$$

$$= \frac{1}{2} \left[\frac{1}{1 - r e^{j\omega_0} z^{-1}} + \frac{1}{1 - r e^{-j\omega_0} z^{-1}} \right] = \frac{1}{2} \left[\frac{z}{z - r e^{j\omega_0}} + \frac{z}{z - r e^{-j\omega_0}} \right], |z| > |r|$$

DIFFERENTIATION

$$x[n] \leftrightarrow X(z) \rightarrow nx[n] \leftrightarrow -z \frac{dX(z)}{dz}$$

$$\frac{d}{dz} z^a = a z^{a-1}$$

PROOF:

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} \rightarrow \frac{dX(z)}{dz} = \frac{d}{dz} \left[\sum_{n=-\infty}^{+\infty} x[n] z^{-n} \right] = \sum_{n=-\infty}^{+\infty} x[n] \frac{dz^{-n}}{dz} = \sum_{n=-\infty}^{+\infty} x[n] (-n) z^{-n-1} = -z^{-1} \sum_{n=-\infty}^{+\infty} nx[n] z^{-n}$$

$$\downarrow$$

$$-z \frac{dX(z)}{dz} = \sum_{n=-\infty}^{+\infty} nx[n] z^{-n} = \mathcal{Z}\{nx[n]\}$$

EX

$$x[n] = n a^n u[n] = ny[n], \quad y[n] := a^n u[n]$$

$$Y(z) = \frac{z}{z-a}, \quad |z| > |a|$$

$$X(z) = -z \frac{dY(z)}{dz} = -z \left[\frac{1}{z-a} - \frac{z}{(z-a)^2} \right] = -z \frac{z-a-z}{(z-a)^2} = -z \frac{-a}{(z-a)^2} = \frac{az^{-1}}{(1-az^{-1})^2}, \quad |z| > |a|$$

TIME REVERSAL

$$x[n] \leftrightarrow X(z), |z| \geq r_x \rightarrow x[-n] \leftrightarrow X\left(\frac{1}{z}\right), |z| \geq \frac{1}{r_x}$$

PROOF:

$$X(z) = \sum_{n=-\infty}^{+\infty} x[-n] z^{-n} = \sum_{\substack{m=-\infty \\ -n=m}}^{+\infty} x[m] z^m = X\left(\frac{1}{z}\right)$$

EX

$$x[n] = a^{-n} u[-n] = y[-n], \quad y[n] := a^n u[n]$$

$$Y(z) = \frac{z}{z-a}, \quad |z| > |a| \rightarrow X(z) = Y\left(\frac{1}{z}\right) = \frac{z^{-1}}{z^{-1}-a}, \quad |z| < \left|\frac{1}{a}\right|$$

$$X(z) = \sum_{n=-\infty}^{+\infty} a^{-n} u[-n] z^{-n} = \sum_{\substack{m=-\infty \\ m=-n}}^{+\infty} a^m u[m] z^m = \sum_{m=0}^{\infty} (az)^m = \frac{1}{1-az}, \quad |az| < 1$$

CONVOLUTION

$$\left. \begin{aligned} x[n] &\longleftrightarrow X(z), \text{ROC: } r_x \\ h[n] &\longleftrightarrow H(z), \text{ROC: } r_h \end{aligned} \right\} y[n] = x[n] * h[n] \longleftrightarrow Y(z) = X(z)H(z), \text{ROC: } r_x \cap r_h$$

PROOF:

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{+\infty} y[n] z^{-n} = \sum_{n=-\infty}^{+\infty} \sum_{m=-\infty}^{+\infty} x[m] h[n-m] z^{-n} = \sum_{m=-\infty}^{+\infty} x[m] \underbrace{\sum_{n=-\infty}^{+\infty} h[n-m] z^{-n}}_{Z\{h[n-m]\}} = \sum_{m=-\infty}^{+\infty} x[m] H(z) z^{-m} = \\ &= H(z) \underbrace{\sum_{m=-\infty}^{+\infty} x[m] z^{-m}}_{Z\{x[m]\}} = H(z) X(z) \quad \# \end{aligned}$$

EX

$$\left. \begin{aligned} x[n] &= 2^n \mu[n] \\ h[n] &= \mu[n] \end{aligned} \right\} y[n] = x[n] * h[n]$$

$$\left. \begin{aligned} X(z) &= \frac{z}{z-2}, \quad |z| > 2 \\ H(z) &= \frac{z}{z-1}, \quad |z| > 1 \end{aligned} \right\} Y(z) = X(z)H(z) = \frac{z^2}{(z-2)(z-1)} = \frac{1}{(1-2z^{-1})(1-z^{-1})} \begin{cases} |z| > 2, & 2 > 1 \\ |z| > 1, & 2 < 1 \end{cases}$$

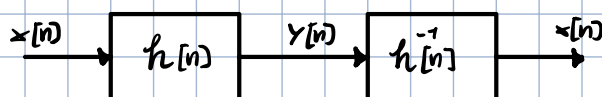
EX

$$x_1[n] = 0.9^n \mu[n] \longleftrightarrow X_1(z) = \frac{z}{z-0.9}, \quad |z| > 0.9$$

$$x_2[n] = \delta[n-2] \longleftrightarrow X_2(z) = z^{-2}, \quad z \neq 0$$

$$x_3[n] = x_1[n] * x_2[n] \longleftrightarrow X_3(z) = \frac{z^{-1}}{z-0.9}, \quad |z| > 0.9$$

INVERSE SYSTEMS



$$x[n] = x[n] * \underbrace{h[n] * h^{-1}[n]}_{\delta[n]} \longrightarrow X(z) = X(z) \underbrace{H(z) H^{-1}(z)}_1$$

$$H(z) H^{-1}(z) = 1 \Rightarrow H^{-1}(z) = \frac{1}{H(z)}$$

WHERE ONE CONVERGES,
THE OTHER EXPLODES

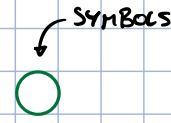
POLES AND ZEROS

IN GENERAL WE CAN EXPRESS THE Z-TRANSFORM AS A FRACTION OF POLYNOMIALS OF z OR z^{-1} . THIS IS CALLED "RATIONAL Z-TRANSFORM"

$$H(z) = \frac{N(z)}{D(z)}$$

$\nearrow$ NUMERATOR
 $\searrow$ DENOMINATOR

THE ROOTS OF $N(z)$ ARE CALLED ZEROS ($N(z)=0 \Rightarrow H(z)=0$)

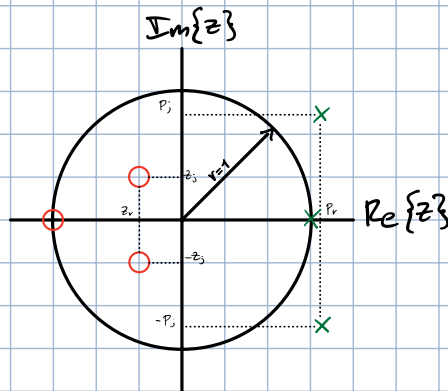


THE ROOTS OF $D(z)$ ARE CALLED POLES ($D(z)=0 \Rightarrow H(z)=\infty$)



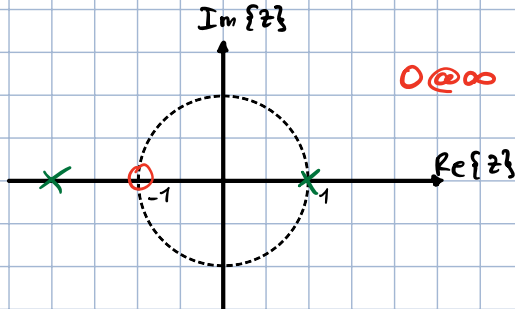
POLES AND ZEROS ARE POINTS IN THE COMPLEX PLANE. THEY COME IN CONJUGATE PAIRS (SYMMETRIC W.R.T. REAL AXIS) EXCEPT FOR REAL POLES AND ZEROS, SINCE THEY LIE EXACTLY ON THE REAL AXIS.

$$\# \text{ POLES } = \# \text{ ZEROS } \checkmark$$



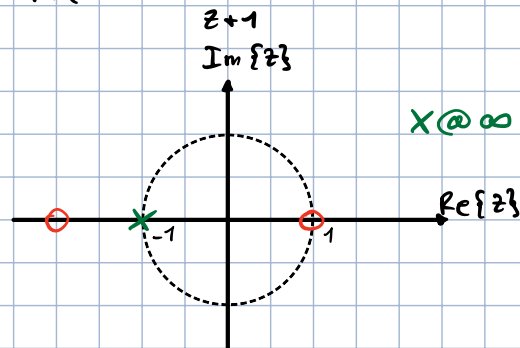
EX

$$H(z) = \frac{z+1}{(z+2)(z-1)}$$



$$\lim_{z \rightarrow \infty} H(z) = \frac{z}{z \cdot z} = \frac{1}{z} = 0$$

$$H^{-1}(z) = \frac{(z+2)(z-1)}{z+1}$$



$$\lim_{z \rightarrow \infty} H^{-1}(z) = \frac{z^2}{z} = z = \infty$$

OBTAINING, N'S JUST AN EXPLANATION OF THESE LIMITS

✓

IN GENERAL WHEN $N(z)$ IS OF ORDER GREATER THAN $D(z)$, THERE IS A POLE IN $z=\infty$. ON THE FLIP SIDE, WHEN $N(z)$ IS OF ORDER SMALLER THAN $D(z)$, THERE IS A ZERO IN $z=\infty$.

INVERSE TRANSFORM

$$x[n] = \frac{1}{2\pi j} \oint_C X(z) z^{n-1} dz$$

WHERE C IS A COUNTERCLOCKWISE CLOSED PATH AROUND THE ORIGIN AND ENTIRELY INSIDE THE ROC.

NOTE THAT IF C IS THE UNIT CIRCLE, THIS IS THE IDFT (GIVEN THE DFT EXISTS IN THE

ROC).

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(e^{j\omega}) e^{j\omega n} d\omega$$

THIS METHOD THOUGH IS WAY TOO COMPLEX. THERE ARE SIMPLER ALTERNATIVES:

- INSPECTION
- POWER SERIES EXPANSION
- PARTIAL FRACTION EXPANSION

INSPECTION

AS HINTED BEFORE, ONE QUICK WAY OF FINDING THE INVERSE TRANSFORM IS RECOGNIZING KNOWN FUNCTIONS. NOTE THAT KNOWING ALSO THE ROC IS FUNDAMENTAL. WITHOUT IT IT'S IMPOSSIBLE TO RETRIEVE THE ORIGINAL SEQUENCE ($\alpha^n u[n]$ AND $\alpha^{-n-1} u[-n-1]$ HAVE SAME $\tilde{x}$, BUT $\neq$ ROC)

EX

$$X(z) = z^{-1} \frac{1}{1 - \frac{1}{4} z^{-1}}, \quad |z| > \frac{1}{4}$$

$$\begin{aligned} &= \underbrace{z^{-1}}_{\delta[n-1]} \underbrace{\frac{z}{z - \frac{1}{4}}}_{\left(\frac{1}{4}\right)^n u[n]} \longleftrightarrow \begin{cases} x[n] = \left(\frac{1}{4}\right)^{n-1} u[n-1], & |z| > \frac{1}{4} \\ x[n] = -\left(\frac{1}{4}\right)^{-n-1} u[-n-1], & |z| < \frac{1}{4} \end{cases} \leftarrow \end{aligned}$$

EX

$$X(z) = \frac{1}{z-1}, \quad |z| > 1 \rightarrow \text{CAUSAL, INFINITE DURATION}$$

$$\begin{aligned} &= \frac{z^{-1}}{1 - z^{-1}} = \underbrace{z^{-1}}_{\delta[n-1]} \cdot \underbrace{\frac{1}{1 - z^{-1}}}_{u[n]} \rightarrow x[n] = \delta[n-1] * u[n] = u[n-1] \end{aligned}$$

POWER SERIES EXPANSION

SUPPOSE $x[n]$ CAUSAL

$$X(z) = \sum_{n=-\infty}^{+\infty} x[n] z^{-n} = \sum_{n=0}^N x[n] z^{-n} = \underline{x[0]} + \underline{x[1]} z^{-1} + \underline{x[2]} z^{-2} + \dots + \underline{x[N]} z^{-N}$$

IN THE CASE OF FINITE SEQUENCES, WE CAN WRITE THE TRANSFORM AS AN (INVERTED) POWER SERIES WHERE THE COEFFICIENTS ARE THEMSELVES THE VALUES OF THE INVERSE TRANSFORM

$$\alpha + \beta z^{-1} + \gamma z^{-2} + \dots \rightarrow [\alpha, \beta, \gamma, \dots]$$

THIS CAN ALSO BE EXTENDED TO INFINITE SERIES USING LONG DIVISION

$$\begin{array}{r|l} N(z) & D(z) \\ & \alpha + \beta z^{-1} + \gamma z^{-2} + \dots \end{array}$$

EX

$$X(z) = z^2 \left(1 - \frac{1}{2}z^{-1}\right)(1+z^{-1})(1-z^{-1}), \quad z \neq 0$$

$$\begin{aligned} & \left| \left(z^2 - \frac{1}{2}z\right) \left(1 - \cancel{z^{-1}} + \cancel{z^{-1}} - z^{-2}\right) = z^2 - 1 - \frac{1}{2}z + \frac{1}{2}z^{-1} = z^2 - \frac{1}{2}z - 1z^0 + \frac{1}{2}z^{-1} \right| \\ & \xrightarrow{S[n+2] \leftrightarrow z^n} S[n+2] - \frac{1}{2}S[n+1] - S[n] + \frac{1}{2}S[n-1] \end{aligned}$$

EX

$$X(z) = (1-2z^{-1})(2-5z^{-1})(3-z^{-1}), \quad z \neq 0$$

$$\left| (2-5z^{-1}-4z^{-1}+10z^{-2})(3-z^{-1}) = (2-9z^{-1}+10z^{-2})(3-z^{-1}) = 6-27z^{-1}+30z^{-2}-2z^{-1}+9z^{-2}-10z^{-3} \right|$$

$$\left| = 6z^0 - 29z^{-1} + 39z^{-2} - 10z^{-3} \rightarrow 6S[n] - 29S[n-1] + 39S[n-2] - 10S[n-3] \right|$$

EX

$$X(z) = \frac{z}{z-1}, \quad |z| > 1 \Rightarrow \text{CAUSAL, INFINITE}$$

THIS DIVIDED BY THIS

$z+0$	$z-1$
$-z+1$	$1+z^{-1}+z^{-2}+\dots$
$\parallel 1$	
$-1+z^{-1}$	
$\parallel z^{-1}$	
$-z^{-1}+z^{-2}$	
$\parallel z^{-2}$	
$\vdots$	

ALL COEFFS ARE 1

- 1) DIVIDE MAX ORDER NUMERATOR BY MAX ORDER DENOMINATOR
- 2) WRITE RESULT
- 3) MULTIPLY WHOLE DENOMINATOR BY PREVIOUSLY FOUND RESULT
- 4) FLIP SIGNS
- 5) ADD
- 6) REPEAT FROM 1) WITH SUM RESULT INSTEAD OF NUMERATOR

$$x[n] = S[n] + S[n-1] + S[n-2] + \dots = u[n]$$

EX

$$X(z) = \frac{z}{z-1}, \quad |z| < 1 \Rightarrow \text{ANTI-CAUSAL, INFINITE}$$

FOR ANTI-CAUSAL SEQUENCES, FLIP ORDER FROM SMALLEST ORDER TO LARGEST
(so 1) BECOMES MAX ORDER / MIN ORDER

z	$-1+z$
$-z+z^2$	$-z-z^2-z^3-z^4-\dots$
$\parallel z^2$	
$-z^2+z^3$	
$\parallel z^3$	
$-z^3+z^4$	
$\parallel z^4$	
$\vdots$	

$x[n] = -S[n+1] - S[n+2] - S[n+3] - \dots = -u[-n-1]$

EX

$$X(z) = \frac{z^2 - 3z + 2}{z^2 - 3z + 2} \quad \text{CAUSAL}$$

$\begin{array}{r} 2z^2 - 3z + 0 \\ -2z^2 + 6z - 4 \\ \hline // \quad 3z - 4 \\ -3z + 9 - 6z^{-1} \\ \hline // \quad 5 - 6z^{-1} \\ -5 + 15z^{-1} - 10z^{-2} \\ \hline // \quad 9z^{-1} - 10z^{-2} \\ -9z^{-1} + 27z^{-2} - 18z^{-3} \\ \hline // \quad 17z^{-2} - 18z^{-3} \\ -17z^{-2} + 51z^{-3} - 34z^{-4} \\ \hline // \quad 33z^{-3} - 34z^{-4} \end{array}$	$\frac{z^2 - 3z + 2}{z^2 - 3z + 2}$ $2 + 3z^{-1} + 5z^{-2} + 9z^{-3} + 17z^{-4} + 33z^{-5} + \dots$ ↓ $x[n] = 2\delta[n] + 3\delta[n-1] + 5\delta[n-2] + 9\delta[n-3] + 17\delta[n-4] + 33\delta[n-5] + \dots$ $[1, 2, 4, 8, 16, 32, \dots] + 1$ $= 2^n u[n] + u[n] = (1 + 2^n) u[n]$
---	--

DOING IT LIKE THIS WORKS BUT IT'S OFTEN COMPLEX. TO MAKE IT EASIER WE CAN SPLIT THE FRACTION IN MULTIPLE SIMPLER ONES AND USE LINEARITY TO RECOMBINE THE SINGLE TRANSFORMS.

$$X(z) = \frac{z^2 - 3z}{z^2 - 3z + 2} = \frac{z^2}{z^2 - 3z + 2} - \frac{3z}{z^2 - 3z + 2}$$

$\begin{array}{r} 2z^2 \\ -2z^2 + 6z - 4 \\ \hline // \quad 6z - 4 \\ -6z + 18 - 12z^{-1} \\ \hline // \quad 14 - 12z^{-1} \\ -14 + 42z^{-1} - 28z^{-2} \\ \hline // \quad 30z^{-1} - 28z^{-2} \end{array}$	$\frac{z^2 - 3z + 2}{2 + 6z^{-1} + 14z^{-2} + 30z^{-3} + \dots}$ ↑ SUPPOSE TO STOP HERE
---	--

$\begin{array}{r} 3z \\ -3z + 9 - 6z^{-1} \\ \hline // \quad 9 - 6z^{-1} \\ -9 + 27z^{-1} - 18z^{-2} \\ \hline // \quad 21z^{-1} - 18z^{-2} \\ -21z^{-1} + 63z^{-2} - 42z^{-3} \\ \hline // \quad 45z^{-2} - 42z^{-3} \end{array}$	$\frac{z^2 - 3z + 2}{3z^{-1} + 9z^{-2} + 21z^{-3} + 45z^{-4} + \dots}$ ↑ SUPPOSE TO STOP HERE
--	--

$$x[n] = 2\delta[n] + 6\delta[n-1] + 14\delta[n-2] + 30\delta[n-3] - 3\delta[n-1] - 9\delta[n-2] - 21\delta[n-3] - 45\delta[n-4]$$

$$= 2\delta[n] + 3\delta[n-1] + 5\delta[n-2] + 9\delta[n-3]$$

$$= (1 + 2^n) u[n]$$

PARTIAL FRACTION EXPANSION

WE KNOW WE CAN EXPRESS A Z TRANSFORM AS A RATIO BETWEEN TWO POLYNOMIALS:

$$X(z) = \frac{\sum_{k=0}^M a_k z^{-k}}{\sum_{k=0}^N b_k z^{-k}}$$

IN SOME CASES WE CAN APPLY LONG DIVISION TO EXPRESS THE TRANSFORM AS A SIMPLE SUM OF POWERS OF Z.

NOW IF $M > N$ WE CAN USE LONG DIVISION TO REWRITE THE RATIO AS:

$$X(z) = \underbrace{\sum_{k=0}^{M-N} A_k z^{-k}}_{\text{RESULT OF LONG DIVISION}} + \underbrace{\frac{\sum_{k=0}^{N-1} a'_k z^{-k}}{\sum_{k=0}^N b_k z^{-k}}}_{\text{REST OF LONG DIVISION}} \quad \leftarrow \text{NOW ORDER}\{N\} = N-1 < \text{ORDER}\{D\} = N$$

ASSUMING WE HAVE DISTINCT POLES, LET'S NOW EXPRES THE DENOMINATOR AS A PRODUCT OF FIRST ORDER TERMS, HIGHLIGHTING THE POLES

$$X(z) = \sum_{k=0}^{M-N} A_k z^{-k} + \frac{\sum_{k=0}^{N-1} a'_k z^{-k}}{b_0 \prod_{k=1}^N (1 - d_k z^{-1})}$$

LET'S NOW DEFINE

$$C(z) := \frac{\sum_{k=0}^{N-1} a'_k z^{-k}}{b_0 \prod_{k=1}^N (1 - d_k z^{-1})} \rightarrow B_k := C(z) (1 - d_k z^{-1}) \Big|_{z=d_k}$$

WE CAN NOW WRITE

$$X(z) = \sum_{k=0}^{M-N} A_k z^{-k} + \sum_{k=1}^N \frac{B_k}{1 - d_k z^{-1}}$$

COEFFICIENTS

THIS IS TRUE IF THE POLES ARE DISTINCT, OTHERWISE THE FORMULATION IS MORE COMPLEX. WE HAVE OBTAINED AN EXPRESSION WHICH IS A LINEAR COMBINATION OF Z TRANSFORMS OF WHICH WE KNOW THEIR INVERSE

$$x[n] = \begin{cases} \sum_{k=0}^{M-N} A_k \delta[n-k] + \sum_{k=1}^N B_k d_k^n u[n] & , \text{ CAUSAL} \\ \sum_{k=0}^{M-N} A_k \delta[n-k] - \sum_{k=1}^N B_k d_k^n u[-n-1] & , \text{ ANTICUSAL} \end{cases}$$

IF IT'S SOMETHING IN BETWEEN, SOME TERMS WILL BE TREATED AS CAUSAL AND SOME AS ANTICUSAL

EX

$$X(z) = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2} z^{-1})(1 - 2 z^{-1})(1 - z^{-1})}, \quad 1/2 < |z| < 2$$

1) CHECK ORDER

$$X(z) = \frac{\sum_{k=0}^M a_k z^{-k}}{\sum_{k=0}^N b_k z^{-k}} \rightarrow M=2, N=3 \rightarrow M < N \Rightarrow \text{NO LONG DIVISION NEEDED}$$

2) REWRITE DENOMINATOR AS PRODUCT OF FIRST ORDER TERMS

→ EXPRESSION IS ALREADY AS DESIRED

3) COMPUTE COEFFICIENTS B_k

$$B_1 = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})(1 - z^{-1})} \cdot \cancel{(1 - \frac{1}{2}z^{-1})} \bigg|_{1 - \frac{1}{2}z^{-1} = 0} = \frac{1 - z^{-1} + z^{-2}}{(1 - 2z^{-1})(1 - z^{-1})} \bigg|_{z^{-1} = 2} = \frac{1 - 2 + 4}{(1 - 2 \cdot 2)(1 - 2)} = \frac{3}{3} = 1$$

$$B_2 = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})\cancel{(1 - 2z^{-1})}(1 - z^{-1})} \cdot \cancel{(1 - 2z^{-1})} \bigg|_{1 - 2z^{-1} = 0} = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - z^{-1})} \bigg|_{z = 2} = \frac{1 - \frac{1}{2} + \frac{1}{4}}{(1 - \frac{1}{4})(1 - \frac{1}{2})} = \frac{\frac{3}{4}}{\frac{3}{4} \cdot \frac{1}{2}} = 2$$

$$B_3 = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})\cancel{(1 - z^{-1})}} \cdot \cancel{(1 - z^{-1})} \bigg|_{1 - z^{-1} = 0} = \frac{1 - z^{-1} + z^{-2}}{(1 - \frac{1}{2}z^{-1})(1 - 2z^{-1})} \bigg|_{z = 1} = \frac{1 - 1 + 1}{(1 - \frac{1}{2})(1 - 2)} = -2$$

4) WRITE EXPRESSION

$$X(z) = \sum_{k=1}^N \frac{B_k}{1 - d_k z^{-1}} = \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{2}{1 - 2z^{-1}} - \frac{2}{1 - z^{-1}} = \frac{z}{z - \frac{1}{2}} + 2 \frac{z}{z - 2} - 2 \frac{z}{z - 1}$$

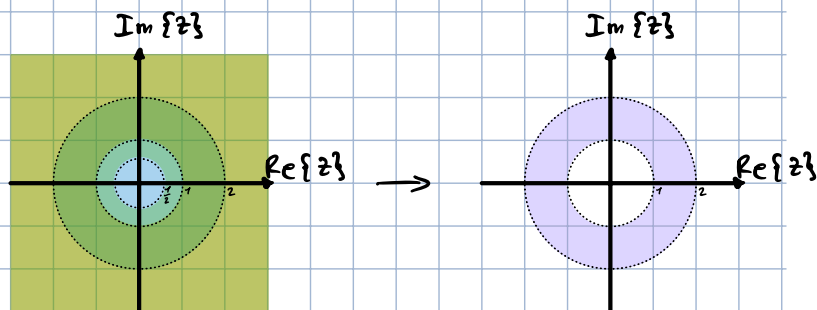
5) CAN'T ANT-TRANSFORM YET, NEED TO CHECK WHETHER TO USE CAUSAL OR ANTICAUSAL FORM. THE GOAL THEN IS TO FIND THE COMBINATION OF ROCs THAT SATISFY THE ORIGINAL ROC.

$$\text{ROC} = \text{ROC}_1 \cap \text{ROC}_2 \cap \text{ROC}_3 : 1 < |z| < 2$$

$$\text{ROC}_1: |z| > \frac{1}{2} \rightarrow \left(\frac{1}{2}\right)^n \mu[n]$$

$$\text{ROC}_2: |z| < 2 \rightarrow -2(2)^n \mu[-n-1]$$

$$\text{ROC}_3: |z| > 1 \rightarrow 2(1)^n \mu[n]$$



6) FINAL EXPRESSION

$$x[n] = \left(\frac{1}{2}\right)^n \mu[n] - 2(2)^n \mu[-n-1] - 2 \mu[n]$$

EX

$$X(z) = \frac{z^3 - 10z^2 - 4z + 4}{2z^3 - 2z^2 - 4z}, \quad |z| < 1, z \neq 0$$

1) EXPRESS AS POWERS OF $-n$

$$\frac{z^3 - 10z^2 - 4z + 4}{2z^3 - 2z^2 - 4z} \cdot \frac{z^{-3}}{z^{-3}} = \frac{1 - 10z^{-1} - 4z^{-2} + 4z^{-3}}{2 - 2z^{-1} - 4z^{-2}}$$

2) CHECK ORDER

$$X(z) = \frac{\sum_{k=0}^M a_k z^{-k}}{\sum_{k=0}^N b_k z^{-k}} \rightarrow M=3, N=2 \rightarrow M < N \Rightarrow \text{LONG DIVISION}$$

3) LONG DIVISION

$$\begin{array}{r|l}
 4z^{-3} - 4z^{-2} - 10z^{-1} + 1 & -4z^{-2} - 2z^{-1} + 2 \\
 -4z^{-3} - 2z^{-2} + 2z^{-1} & -z^{-1} + \frac{3}{2} \\
 \hline
 // -6z^{-2} - 8z^{-1} + 1 & \\
 6z^{-2} + 3z^{-1} - 3 & \\
 \hline
 // -5z^{-1} - 2 &
 \end{array}
 \rightarrow X(z) = -z^{-1} + \frac{3}{2} + \frac{-5z^{-1} - 2}{-4z^{-2} - 2z^{-1} + 2}$$

4) DENOMINATOR AS PRODUCT OF POLYNOMIALS OF ORDER 1, AKA FIND ZEROS

$$-4z^{-2} - 2z^{-1} + 2 = 0 \rightarrow -4 - 2z + 2z^2 = 0 \rightarrow z(z^2 - z - 2) = 0 \rightarrow z^2 - z - 2 = 0$$

$$z_{1,2} = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} < \frac{-1}{2} \rightarrow (1+z^{-1})(1-2z^{-1})$$

$$X(z) = -z^{-1} + \frac{3}{2} + \frac{-5z^{-1} - 2}{2(1+z^{-1})(1-2z^{-1})}$$

REMEMBER THIS

5) COMPUTE COEFFICIENTS FOR PARTIAL FRACTION EXPANSION

$$B_1 = \frac{-5z^{-1} - 2}{2(1-2z^{-1})} \bigg|_{z^{-1} = -1} = \frac{-5-2}{2(1+2)} = \frac{-7}{6} = -\frac{1}{2}$$

$$B_2 = \frac{-5z^{-1} - 2}{2(1+z^{-1})} \bigg|_{z^{-1} = \frac{1}{2}} = \frac{-5-\frac{5}{2}}{2(1+\frac{1}{2})} = -\frac{15}{6} = -\frac{5}{2}$$

$$X(z) = -z^{-1} + \frac{3}{2} + \frac{1/2}{1+z^{-1}} - \frac{3/2}{1-2z^{-1}}$$

6) FIND ROCs

$$|z| < 1, z \neq 0$$

$$\text{ROC}_1: z \neq 0 \rightarrow -\delta[n-1] + \frac{3}{2} \delta[n]$$

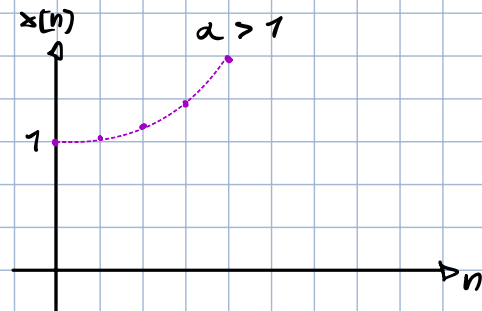
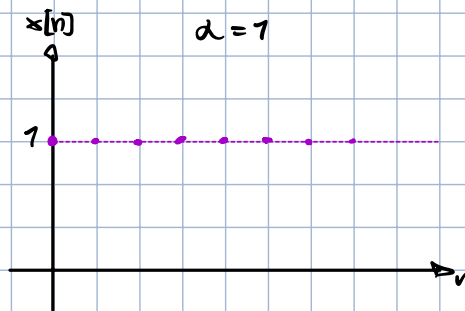
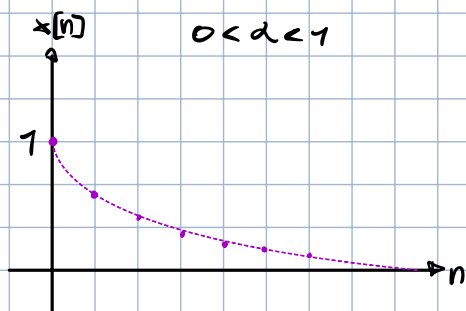
$$\text{ROC}_2: |z| < 1 \rightarrow \frac{1}{2} u[n-1]$$

$$\text{ROC}_3: |z| < 2 \rightarrow -\frac{3}{2} (2)^n u[n-1]$$

$$x[n] = -\delta[n-1] + \frac{3}{2} \delta[n] + \frac{1}{2} u[n-1] - \frac{3}{2} (2)^n u[n-1]$$

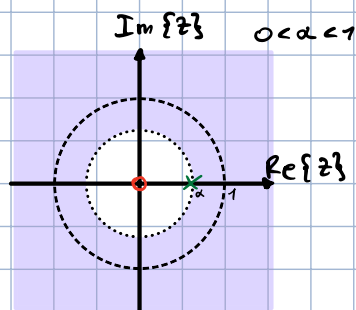
EX

$$x[n] = \alpha^n u[n]$$

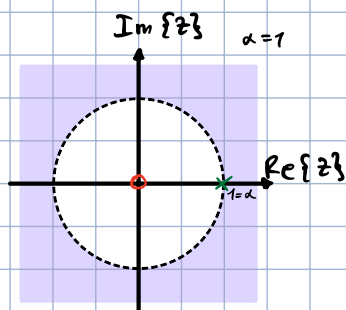


$$X(z) = \frac{1}{1 - \alpha z^{-1}} = \frac{z}{z - \alpha} \rightarrow \text{ORDER: 1} \Rightarrow 1 \text{ POLE} + 1 \text{ ZERO}$$

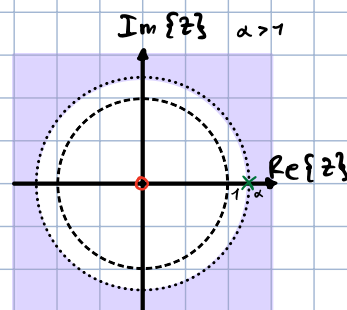
NO POLE IN ∞ , BECAUSE
 $\lim_{z \rightarrow \infty} \frac{z}{z - \alpha} = \frac{z}{z} = 1 \Rightarrow \text{SAME ORDER}$



UNIT CIRCLE IS INCLUDED



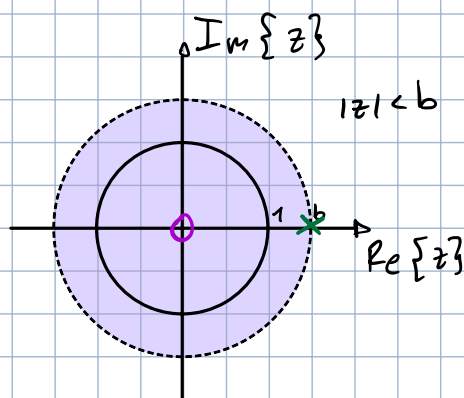
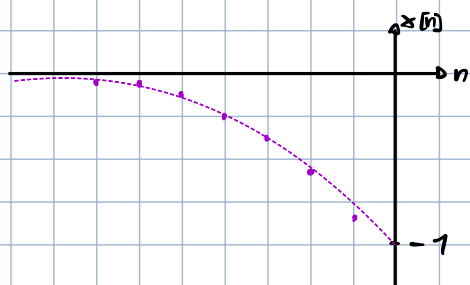
UNIT CIRCLE IS EXCLUDED



UNIT CIRCLE IS EXCLUDED

EX

$$y[n] = -b^n u[-n-1], \quad b > 1 \rightarrow Y(z) = \frac{z}{z - b}, \quad \text{ANTICAUSAL} \Rightarrow |z| < b$$



EX

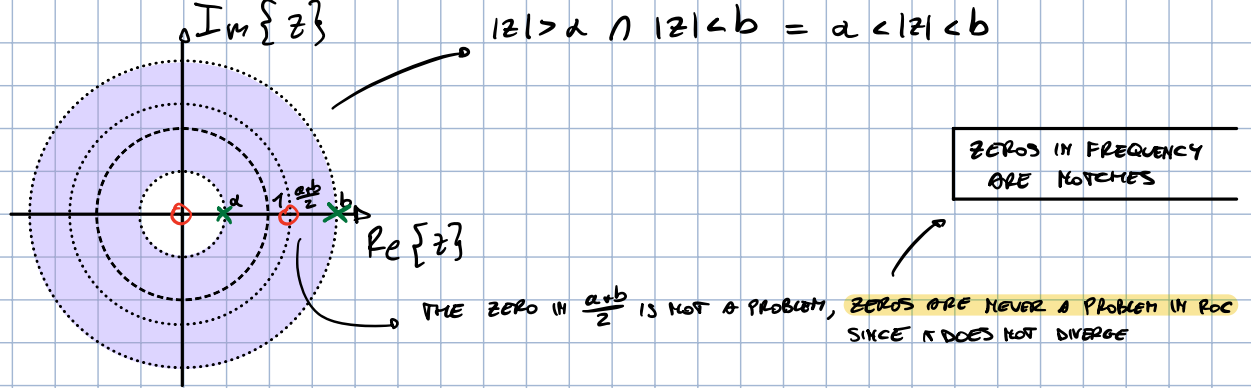
$$x[n] + y[n] = \alpha^n u[n] - b^n u[-n-1] \rightarrow \frac{z}{z - \alpha} + \frac{z}{z - b}, \quad 0 < \alpha < 1, \quad b > 1$$

CAUSAL ANTICAUSAL

$$\frac{z}{z - \alpha} + \frac{z}{z - b} = z \left(\frac{1}{z - \alpha} + \frac{1}{z - b} \right) = z \frac{z - b + z - \alpha}{(z - \alpha)(z - b)} = z \frac{2z - b - \alpha}{(z - \alpha)(z - b)} = \frac{2z(z - \frac{\alpha+b}{2})}{(z - \alpha)(z - b)}$$

NEW ZERO!

POLES: α, b
 ZEROS: $0, \frac{\alpha+b}{2}$



AT THE POLES THE FUNCTION EXPLODES, AT THE ZEROS IT GOES TO ZERO $\Rightarrow$ CLOSE TO POLES IT AMPLIFIES, CLOSE TO ZEROS IT ATTENUATES.

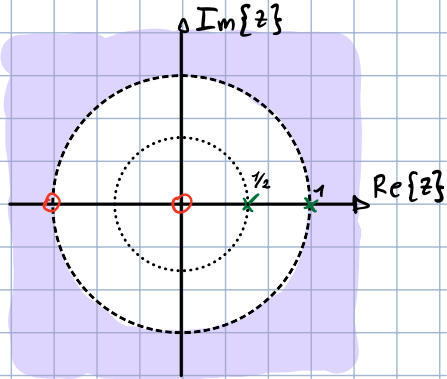
EX

$$X(z) = \frac{1+z^{-1}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})} = \frac{z^2}{z^2} \frac{1+z^{-1}}{(1-z^{-1})(1-\frac{1}{2}z^{-1})} = \frac{z(z+1)}{(z-1)(z-\frac{1}{2})}$$

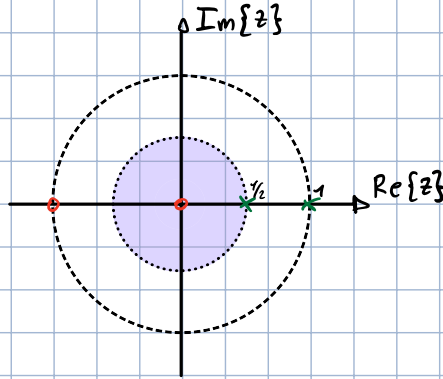
POLES: $1, \frac{1}{2}$
ZEROS: $0, -1$

No pole at ∞
Because $\lim_{z \rightarrow \infty} \frac{z}{z} = 1$
 $\Rightarrow$ SAME ORDER $M=N$

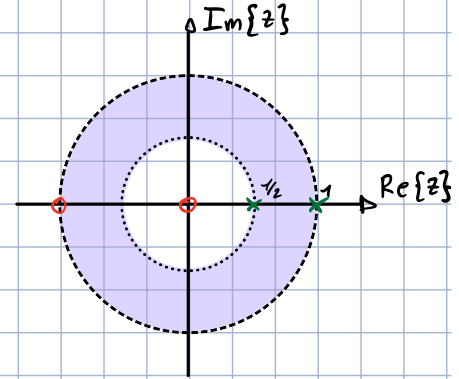
THREE POSSIBLE ROCs:



CAUSAL
 $|z| > 1$



ANTICAUSAL
 $|z| < \frac{1}{2}$



TWO-SIDED
 $\frac{1}{2} < |z| < 1$